1) a)

Stretch parallel to the x-axis scale factor 2 and stretch parallel to the y-axis scale factor 3

1) b)

$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2-\lambda & 0 \\ 0 & 3-\lambda \end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (2-\lambda)(3-\lambda)$

So $(2-\lambda)(3-\lambda) = 0$

$\therefore \lambda = 2$ or $\lambda = 3$

When $\lambda_1 = 2$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow y = 0$ for any value of x so $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

When $\lambda_2 = 3$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$

$\begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -x = 0$ for any value of y so $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\therefore \lambda_1 = 2$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\lambda_2 = 3$, $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

1) c)

$y = 0$ and $x = 0$

1) d)

$\det \mathbf{A} = 6$ so the area of the image has been enlargened by a factor of 6

2) a)

Reflection across the line $y = x$

2) b)

$\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = \lambda\begin{pmatrix}x\\ y\end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} - \lambda\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = \begin{pmatrix}-\lambda & 1\\ 1 & -\lambda\end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (-\lambda)(-\lambda) - 1$

So $\lambda^2 - 1 = 0$

$\therefore \lambda = -1$ or $\lambda = 1$

When $\lambda_1 = -1$ then $\mathbf{A} + \mathbf{I} = \begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$

$\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = 0 \Rightarrow x = -y$ so $\mathbf{v}_1 = \begin{pmatrix}1\\ -1\end{pmatrix}$

When $\lambda_2 = 1$ then $\mathbf{A} - 1\mathbf{I} = \begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix}$

$\begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = 0 \Rightarrow -x + y = 0$ so $\mathbf{v}_2 = \begin{pmatrix}1\\ 1\end{pmatrix}$

$\therefore \lambda_1 = -1$, $\mathbf{v}_1 = \begin{pmatrix}1\\ -1\end{pmatrix}$ and $\lambda_2 = 1$, $\mathbf{v}_2 = \begin{pmatrix}1\\ 1\end{pmatrix}$

2) c)

$y = -x$ and $y = x$

2) d)

$\det\mathbf{A} = -1$ so the area is preserved, but the orientation is reversed.

3) a)

Rotation $90°$ anti-clockwise about the origin.

3) b)

$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = \lambda\begin{pmatrix}x\\ y\end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix} - \lambda\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = \begin{pmatrix}-\lambda & -1\\ 1 & -\lambda\end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (-\lambda)(-\lambda) + 1$

So $\lambda^2 + 1 = 0$

$\therefore \lambda = -\mathrm{i}$ or $\lambda = \mathrm{i}$

3) c)

Since the eigenvalues are complex, there are no invariant lines.

Geometrically, since every line rotates, no real line maps onto itself.

3) d)

$\det \mathbf{A} = 1$ so the area and orientation is preserved.

4) a)

$\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2-\lambda & 0 \\ 0 & -2-\lambda \end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (2-\lambda)(-2-\lambda)$

So $(2-\lambda)(-2-\lambda) = 0$

$\therefore \lambda = -2$ or $\lambda = 2$

When $\lambda_1 = -2$ then $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow 4x = 0$ so $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

When $\lambda_2 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 0 \\ 0 & -4 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 \\ 0 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -4y = 0$ so $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\therefore \lambda_1 = -2$, $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\lambda_2 = 2$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

4) b)

$y = 0$ and $x = 0$

4) c)

$\det \mathbf{A} = -4$ so the area is enlarged by factor of 4, and the orientation is reversed.

4) d)

The eigenvalues and eigenvectors tell us:

- A stretch of scale factor 2 in the direction $\begin{pmatrix}0\\1\end{pmatrix}$
- A stretch of scale factor 2 in the direction $\begin{pmatrix}1\\0\end{pmatrix}$ with a reflection in the x-axis

$\therefore$ Enlargement of scale factor 2 and a reflection in the x-axis.

5) a)

$$\begin{pmatrix}\frac{13}{5} & \frac{6}{5}\\ \frac{6}{5} & \frac{22}{5}\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\lambda\begin{pmatrix}x\\y\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}\frac{13}{5} & \frac{6}{5}\\ \frac{6}{5} & \frac{22}{5}\end{pmatrix}-\lambda\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}=\begin{pmatrix}\frac{13}{5}-\lambda & \frac{6}{5}\\ \frac{6}{5} & \frac{22}{5}-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=\left(\tfrac{13}{5}-\lambda\right)\left(\tfrac{22}{5}-\lambda\right)-\tfrac{36}{25}$$

So $\lambda^2-7\lambda+10=(\lambda-2)(\lambda-5)=0$

$\therefore \lambda=2$ or $\lambda=5$

5) b)

When $\lambda_1=2$ then $\mathbf{A}-2\mathbf{I}=\begin{pmatrix}\frac{3}{5} & \frac{6}{5}\\ \frac{6}{5} & \frac{12}{5}\end{pmatrix}$

$$\begin{pmatrix}\frac{3}{5} & \frac{6}{5}\\ \frac{6}{5} & \frac{12}{5}\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=0\Rightarrow\tfrac{3}{5}x+\tfrac{6}{5}y=0 \text{ so } y=-\tfrac{1}{2}x$$

When $\lambda_2=5$ then $\mathbf{A}-5\mathbf{I}=\begin{pmatrix}-\frac{12}{5} & \frac{6}{5}\\ \frac{6}{5} & -\frac{3}{5}\end{pmatrix}$

$$\begin{pmatrix}-\frac{12}{5} & \frac{6}{5}\\ \frac{6}{5} & -\frac{3}{5}\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=0\Rightarrow-\tfrac{12}{5}x+\tfrac{6}{5}y=0 \text{ so } y=2x$$

$\therefore y=-\frac{1}{2}x$ and $y=2x$

5) c)

The eigenvalues and eigenvectors tell us:

- A stretch of scale factor 2 in the direction $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$
- A stretch of scale factor 5 in the direction $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

They are perpendicular so there are no rotations or shears.

$\therefore$ A stretch of scale factor 2 along $y = -\frac{1}{2}x$ and a stretch of scale factor 5 along $y = 2x$

6) a)

Since $2\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ then $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is an eigenvector.

6) b)

$\lambda = 2$

7) a)

Since $-2\begin{pmatrix} 2 \\ -5 \end{pmatrix} = \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ then $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$ is an eigenvector.

7) b)

$\lambda = -2$

8) a)

$\begin{pmatrix} p & q \\ q & p \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} p & q \\ q & p \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p-\lambda & q \\ q & p-\lambda \end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (p-\lambda)(p-\lambda) - q^2$

So $\lambda^2 - 2\lambda p + p^2 - q^2 = 0$

$(\lambda - (p-q))(\lambda - (p+q)) = 0$

$\therefore \lambda = p - q$ or $\lambda = p + q$

When $\lambda_1 = p - q$ then $\mathbf{A} - (p-q)\mathbf{I} = \begin{pmatrix} q & q \\ q & q \end{pmatrix}$

$\begin{pmatrix} q & q \\ q & q \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow qx + qy = 0$ so $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

When $\lambda_2 = p + q$ then $\mathbf{A} - (p+q)\mathbf{I} = \begin{pmatrix} -q & q \\ q & -q \end{pmatrix}$

$\begin{pmatrix} -q & q \\ q & -q \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -qx + qy = 0$ so $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\therefore \lambda_1 = p - q$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\lambda_2 = p + q$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

8) b)

Invariant lines are $y = -x$ and $y = x$

Consider gradients: $-1 \times 1 = -1$ so the invariant lines are perpendicular.

8) c)

The eigenvalues and eigenvectors tell us:

- A stretch of scale factor $p - q$ in the direction $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$
- A stretch of scale factor $p + q$ in the direction $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

They are perpendicular so there are no rotations or shears.

$\therefore$ A stretch of scale factor $p - q$ along $y = -x$ and a stretch of scale factor $p + q$ along $y = x$

9) a)

$\begin{pmatrix}3 & 1\\ 1 & 3\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = \lambda\begin{pmatrix}x\\ y\end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix}3 & 1\\ 1 & 3\end{pmatrix} - \lambda\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = \begin{pmatrix}3-\lambda & 1\\ 1 & 3-\lambda\end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (3-\lambda)(3-\lambda) - 1$

So $\lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2) = 0$

$\therefore \lambda = 2$ or $\lambda = 4$

9) b)

When $\lambda_1 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$

$\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = 0 \Rightarrow x + y = 0$ so $y = -x$

When $\lambda_2 = 4$ then $\mathbf{A} - 4\mathbf{I} = \begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix}$

$\begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = 0 \Rightarrow -x + y = 0$ so $y = x$

$\therefore y = -x$ and $y = x$

9) c)

$\mathbf{A}^2$ has eigenvalues 4 and 16

- A stretch of scale factor 4 in the direction $\begin{pmatrix}1\\ -1\end{pmatrix}$
- A stretch of scale factor 16 in the direction $\begin{pmatrix}1\\ 1\end{pmatrix}$

They are perpendicular so there are no rotations or shears.

So $\mathbf{A}^2$ stretches along the line $y = -x$ by scale factor 4 and along the line $y = x$ by scale factor 16

10)

$\begin{pmatrix}-10 & 8\\ -16 & 14\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = \lambda\begin{pmatrix}x\\ y\end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix}-10 & 8\\ -16 & 14\end{pmatrix} - \lambda\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = \begin{pmatrix}-10-\lambda & 8\\ -16 & 14-\lambda\end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (-10-\lambda)(14-\lambda) + 128$

So $\lambda^2 - 4\lambda - 12 = (\lambda + 2)(\lambda - 6) = 0$

$\therefore \lambda = -2$ or $\lambda = 6$

When $\lambda_1 = -2$ then $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} -8 & 8 \\ -16 & 16 \end{pmatrix}$

$\begin{pmatrix} -8 & 8 \\ -16 & 16 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -8x + 8y = 0$ so $x = y$

When $\lambda_2 = 6$ then $\mathbf{A} - 6\mathbf{I} = \begin{pmatrix} -16 & 8 \\ -16 & 8 \end{pmatrix}$

$\begin{pmatrix} -16 & 8 \\ -16 & 8 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -16x + 8y = 0$ so $y = 2x$

$\therefore y = x$ and $y = 2x$

$\mathbf{A}^3$ has eigenvalues $(-2)^3 = -8$ and $6^3 = 216$

- A stretch of scale factor -8 in the direction $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- A stretch of scale factor 216 in the direction $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

So $\mathbf{A}^3$ stretches along the line $y = x$ by scale factor 8, with a change in orientation, and along the line $y = 2x$ by scale factor 216 with orientation preserved.